

Efficient Deep Learning and Spatial Modeling for Mapping Tropical Forest Canopies

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Self-Introduction

- **Positions:**

Research Assistant Professor, Department of Computer Science;
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- **Research:**

Efficient machine learning; statistical & spatial modeling; interpretable deep learning (applications: remote sensing, ecology, healthcare).

- **Education:**

Ph.D. in Mathematics, City University of Hong Kong.

B.Sc. in Computing Mathematics, City University of Hong Kong.

Outline

- 1 Motivation and Problem
- 2 The PALMS Dataset
- 3 The PRISM Framework
- 4 Spatial Distribution Modeling
- 5 Conclusion

Motivation and Problem

The Importance of Palms in Tropical Forests



(a) Cases from existing studies

(b) Cases from our dataset

Figure: Comparative Samples of Manual Labels.

Ecological & Economic Significance of Palms:

- Vital to tropical forest ecology, biodiversity, and conservation planning [1].
- Support sustainable livelihoods and are key resources for tropical wildlife [2].
- Can serve as bioindicators of forest health and environmental impact.

Problem: Most prior work targets plantations; *natural forests* show irregular spacing, overlapping crowns, complex backgrounds, uneven lighting.

Research Challenges & Our Contributions

Key Challenges

- **Image Variability:** Occlusion from overlapping canopies and inconsistent lighting degrade image quality.
- **Data Scarcity:** High-quality annotated datasets for tropical forests are rare and difficult to create.
- **Spatial Analysis Gap:** Most work focuses only on detection, not the large-scale spatial structure of populations.

Our Contributions

- 1 **New Dataset (PALMS):** Including 21 forest sites' orthomosaics in western Ecuador, with over 8,800 bounding boxes and 5,000 palm center annotations.
- 2 **PRISM Framework:** A flexible, interpretable pipeline for palm detection, segmentation, and counting.
- 3 **Spatial Modeling:** A Poisson-Gaussian model that simulates and provides insight into the ecological processes driving palm distribution.

The PALMS Dataset

The PALMS Dataset: Data from the Field

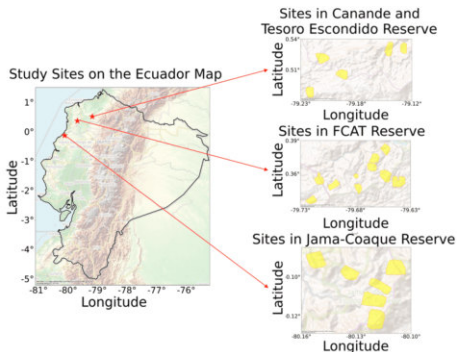


Figure: Study sites across a rainfall and ecological gradient in western Ecuador.

- Data collected from four reserves spanning wet to dry tropical forests.
- Captures high variation in palm species, density, and canopy structure.
- High-resolution orthomosaics created from thousands of UAV images.



Figure: Bounding box annotations.



Figure: Georeferenced palm centers.

The PRISM Framework

The PRISM Framework at a Glance

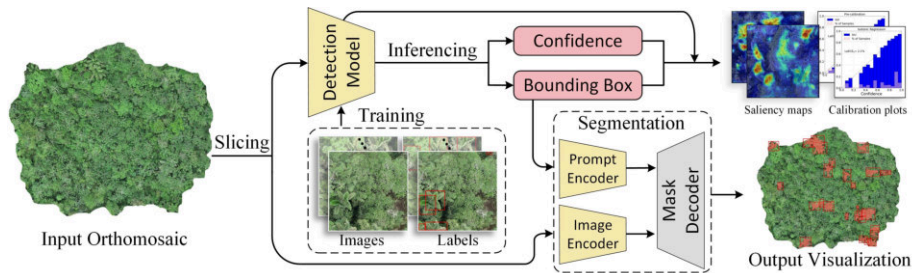


Figure: The PRISM Pipeline: From detection to segmentation and analysis [3].

Core Components:

- **Detection:** Fine-tuned models locate palms in orthomosaic patches.
- **Segmentation:** Detections are used as prompts for a zero-shot Segment Anything Model (SAM) to generate precise masks.
- **Mapping:** Outputs are georeferenced for landscape-scale analysis.
- **Interpretability:** Grad-CAM and calibration analysis enhance model reliability.

Detection and Segmentation Performance

Table: Detection model performance comparison.

Model	GFLOPS ↓	Params (M) ↓	FPS ↑	Precision ↑	Recall ↑	AP ₅₀ ↑	AP ₇₅ ↑	mAP ↑
DINO	1920.3	218.2	18.98 ± 0.95	0.7629 ± 0.0177	0.8494 ± 0.0071	0.8169 ± 0.0166	0.5455 ± 0.0150	0.5102 ± 0.0101
DDQ	1232.6	218.6	19.18 ± 0.96	0.7825 ± 0.0124	0.8566 ± 0.0123	0.8541 ± 0.0129	0.6354 ± 0.0137	0.5736 ± 0.0130
RT-DETR	222.5	65.5	151.49 ± 0.70	0.8869 ± 0.0230	0.7598 ± 0.0310	0.8416 ± 0.0181	0.6198 ± 0.0181	0.5769 ± 0.0145
YOLOv8	226.7	61.6	174.92 ± 0.86	0.8729 ± 0.0165	0.7997 ± 0.0203	0.8667 ± 0.0141	0.6777 ± 0.0137	0.6148 ± 0.0128
YOLOv9	169.5	53.2	114.96 ± 0.30	0.8763 ± 0.0176	0.7976 ± 0.0209	0.8741 ± 0.0109	0.6762 ± 0.0146	0.6162 ± 0.0122
YOLOv10	169.8	31.6	177.04 ± 1.14	0.8716 ± 0.0121	0.7968 ± 0.0089	0.8626 ± 0.0129	0.6794 ± 0.0112	0.6173 ± 0.0090
YOLO11	194.4	56.8	170.40 ± 0.95	0.8721 ± 0.0095	0.7896 ± 0.0127	0.8684 ± 0.0108	0.6677 ± 0.0180	0.6115 ± 0.0109

Key Findings:

- **YOLOv10 (Selected):** Best overall trade-off, achieving the highest mAP, AP₇₅ and inference speed with the fewest parameters.
- **DDQ:** Highest recall, ideal when finding all possible instances is prioritized.
- **RT-DETR:** Highest precision, but misses more palms (lower recall).

Segmentation Performance: Comparing SAM Variants

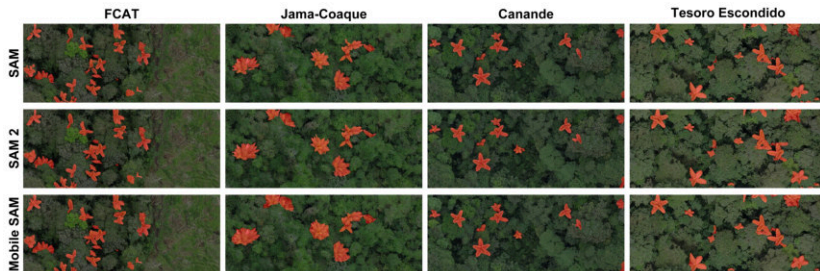


Figure: Visual comparison of SAM variants for zero-shot palm segmentation.

Key Findings

- We use the detector's bounding boxes as prompts for **zero-shot segmentation**.
- A comparison revealed distinct behaviors on our dataset:
 - **Original SAM:** Occasionally produces incomplete segments (under-segments).
 - **MobileSAM:** Tends to over-segment into non-palm areas.
 - **SAM 2 (Selected):** Provides the most balanced and accurate segmentation.

Visualizing What the Model "Sees" with Grad-CAM

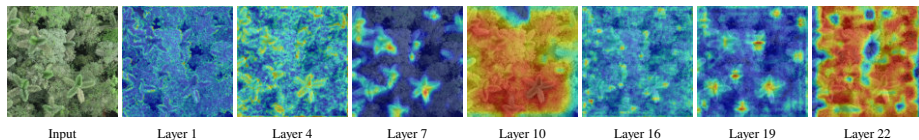


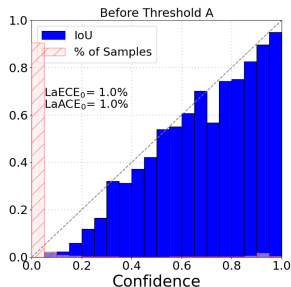
Figure: Hierarchical Feature Learning in YOLOv10 through Grad-CAM Visualizations.

Hierarchical Feature Learning

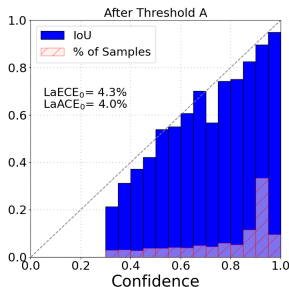
The analysis confirms the model learns a meaningful progression: early layers focus on low-level edges and textures; intermediate layers integrate spatial context; and deep layers exhibit focused activation over entire palm crowns.

Model Interpretability: A Step-by-Step Guide to Calibration

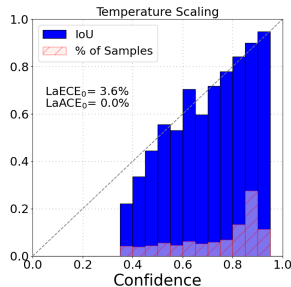
1. Initial Output



2. LRP Thresholding



3. Post-Hoc Calibration



The Problem

The uncalibrated model is unreliable; its confidence scores are poorly correlated with true accuracy (IoU).

The First Fix

LRP-based thresholding is first applied to prune the large number of unreliable, low-confidence predictions.

The Final Result

A post-hoc method (e.g., Temperature Scaling) is then applied to align confidence with accuracy.

Quantitative Analysis of Counting Performance

Table: Counting performance across four distinct ecological sites.

Site	Area (ha)	Counts	Pred2GT		GT2Pred	
			Ratio	Median (m)	Ratio	Median (m)
FCAT	21.62	471	0.9361	1.10	0.8854	0.77
Jama-Coaque	111.93	952	0.9348	1.50	0.8151	1.14
Canandé	101.20	1,273	0.8956	0.82	0.7667	0.72
Tesoro Escondido	86.76	2,330	0.8981	1.09	0.9253	0.91

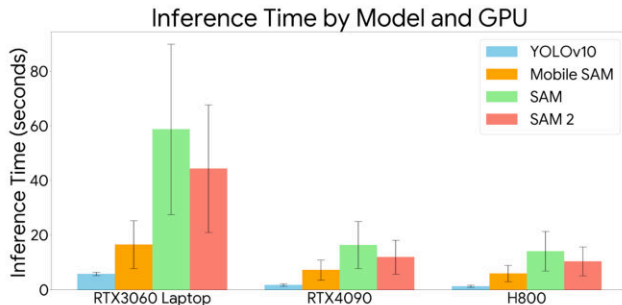
Key Metrics:

- **Pred2GT Ratio (Precision):** Proportion of predictions matched to a ground truth palm.
- **GT2Pred Ratio (Recall):** Proportion of ground truth palms matched by a prediction.

Summary of Findings:

- Precision is high across all sites, indicating the model generates few false positives.
- Recall is more variable, showing that detecting every true palm is harder and site-dependent.
- Sites like Tesoro Escondido show balanced performance, while Canandé reveals recall limitations (some palms are missed).

Feasibility for Real-Time Analysis



Key Findings:

- **Detection is Real-Time Ready:** YOLOv10 is fast enough (1.2–5.7s / image) for live processing on a UAV, even with mid-range hardware.
- **Segmentation is Costly:** Segmentation speed varies greatly, making it an optional step for time-critical missions.
- **Conclusion:** The core detection pipeline is efficient and stable, meeting the requirements for field deployment.

Spatial Distribution Modeling

Background: Measuring Spatial Point Patterns

Goal

To quantify if a spatial point pattern is clustered, random, or regular by comparing it against a model of Complete Spatial Randomness (CSR).

Ripley's G & F Functions

These are cumulative distribution functions (CDFs) that measure nearest-neighbor distances at a given distance radius d :

- $G(d)$: CDF of distances from each point to its *nearest neighbor in the pattern*. It quantifies internal clustering.
- $F(d)$: CDF of distances from *random locations* to the nearest point in the pattern. It quantifies empty space.

Formal Definitions

Function	Formulation
$G(d)$	$\frac{1}{N_o} \sum_{i=1}^{N_o} \mathbb{1}(\hat{d}_i < d)$
$F(d)$	$\frac{1}{N_r} \sum_{j=1}^{N_r} \mathbb{1}(\tilde{d}_j < d)$
$J(d)$	$\frac{1-G(d)}{1-F(d)}$

Here, N_o is the number of observed points and $\hat{d}_i$ is the distance from point i to its nearest neighbor. N_r is the number of random points and $\tilde{d}_j$ is the distance from random point j to the nearest observed point.

Modeling Palm Distributions

Our Goal

To simulate palm spatial patterns that match observed distributions and to understand the ecological drivers of reproduction (e.g., long-range dispersal vs. local clustering).

The Core Mechanism: A Hybrid Generative Process

The model simulates palm propagation by combining two key ecological processes, controlled by two interpretable parameters [4]:

- **Global Dispersal (Poisson):** With probability $(1 - p)$, a new palm is placed randomly, representing animal-mediated or long-range dispersal.
- **Local Clustering (Gaussian):** With probability p , a new palm is placed near a parent, drawn from $\mathcal{N}(\mathbf{x}_{\text{parent}}, \sigma^2 \mathbf{I})$, representing local seed drop.

Parameter Fitting

The optimal parameters (p^*, σ^*) are found by identifying the pair that generates simulated patterns whose spatial statistics (Ripley's G and F functions) most closely match those of the observed data.

The Poisson-Gaussian Algorithm: Implementation Details

Algorithm Pseudocode

Input: Candidate params $(\mathbf{p}, \sigma)$, Observed points X

Output: Optimal params (p^*, σ^*)

Initialize: $d_{min} \leftarrow \infty$

Pre-compute: Observed Ripley's stats G_{obs}, F_{obs} .

for each (p, σ) in grid **do**

$d_{total} \leftarrow 0$

for $i = 1$ **to** N simulations **do**

 1. Generate simulated set $\hat{X}_i$ via the

 Poisson-Gaussian process.

 2. Compute simulated stats G_{sim}, F_{sim} .

 3. Calculate discrepancy d_i .

 4. Add d_i to d_{total} .

end for

if $d_{total} < d_{min}$ **then**

 Update $d_{min}, p^* \leftarrow p, \sigma^* \leftarrow \sigma$.

end if

end for

return p^*, σ^*

Discrepancy Metric

The discrepancy d_i for each simulation is the integrated absolute difference between observed and simulated Ripley's functions:

$$d_i = \int |\mathbf{g}_{obs} - \mathbf{g}_{sim}| + \int |\mathbf{f}_{obs} - \mathbf{f}_{sim}|$$

Optimization Process

A grid search is performed over the parameter space. The pair (p^*, σ^*) that minimizes the total discrepancy over all simulations is selected as the optimal fit.

Observed Spatial Patterns: Clustered or Random?

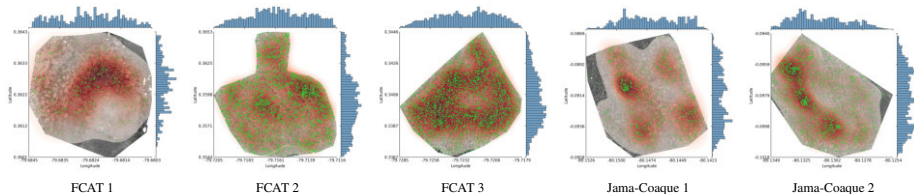


Figure: Kernel Density Estimates (KDEs) of detected palm locations across five study sites, visually suggesting non-random clustering.

The Central Research Question

The spatial arrangement of palms appears clustered, but is this pattern statistically significant? We test the observed distributions against a null model of CSR.

Statistical Proof: Analysis with Ripley's Functions

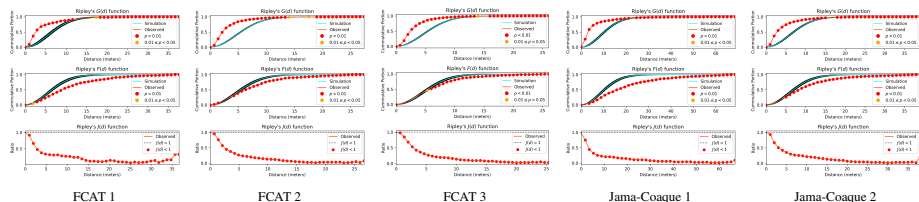


Figure: Ripley's function plots for each site. The observed pattern (red curve) is compared against the 95% confidence envelope of a CSR process (blue curve with shaded area).

Conclusion from the Analysis

The results confirm a **statistically significant** departure from randomness across all sites, with Ripley's functions revealing both dense internal clustering (*G*-function) and large empty spaces (*F*-function). This strong, non-random aggregation justifies our development of a more complex reproduction model.

Simulation Results: Replicating Observed Patterns

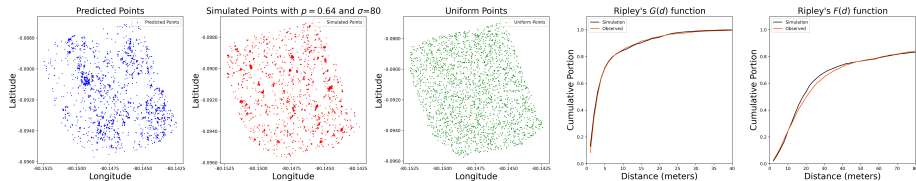


Figure: Visual and statistical comparison for the site Jama-Coaque 1. From left to right: PRISM prediction, model simulation, random distribution, G -, and F -function comparison.

Optimal Parameters

Fitted (p^* , σ^*) across sites.

Site	p^*	σ^*
FCAT 1	0.49	50
FCAT 2	0.52	70
FCAT 3	0.46	70
Jama-Coaque 1	0.64	80
Jama-Coaque 2	0.51	60

Key Findings

- The optimal parameters are highly consistent.
- This indicates a stable balance between local clustering (within a ~ 2 -4 meter radius) and random, long-range dispersal.
- The strong alignment between simulated and observed Ripley's functions (right panels) validates the model's fidelity.

Conclusion

Conclusion

Summary of Contributions

- Created **PALMS**, a new, large-scale annotated dataset for palm detection in ecologically diverse tropical forests.
- Developed **PRISM**, an end-to-end framework for efficient palm detection, segmentation, and counting from UAV imagery.
- Introduced a simple, two-parameter **Poisson-Gaussian model** that successfully replicates the complex spatial dynamics of palm distribution, as validated by Ripley's functions.

Future Directions

- **Dataset Expansion:** A 1000 km² Amazonian region in Peru.
- **Model Enhancement:** Reduce supervision requirements and enable robust application on lower spatial-resolution imagery.
- **Deployment & Extension:** Real-time, on-device deployment (e.g., NVIDIA Jetson) and extension to species-level classification.

References I

- [1] W. L. Eiserhardt et al. “Geographical ecology of the palms (Arecaceae): determinants of diversity and distributions across spatial scales”. In: *Ann Bot* 108.8 (2011), pp. 1391–1416.
- [2] N. C. Pitman et al. “Distribution and abundance of tree species in swamp forests of Amazonian Ecuador”. In: *Ecography* 37.9 (2014), pp. 902–915.
- [3] K. Cui et al. “Detection and Geographic Localization of Natural Objects in the Wild: A Case Study on Palms”. In: *Proc. IJCAI*. Accepted, to appear. 2025.
- [4] K. Cui et al. “Efficient Localization and Spatial Distribution Modeling of Canopy Palms Using UAV Imagery”. In: *IEEE Transactions on Geoscience and Remote Sensing* (2025).

Thank You

Questions & Discussion

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